

Restriction Categories as Double Categories

V. Jelínek¹ T. Tsantilas²

¹Department of Informatics,
University of Sussex

²School of Applied Mathematical and Physical Sciences,
National Technical University of Athens

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mentors: N. Arkor, B. Clarke

group: Y. Chamoun, V. Jelínek, H. Nasu, T. Tsantilas

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The goal of this talk is to introduce restriction categories and double categories and compare them. For the comparison, we choose the notion of product.

1 Restriction Categories

2 Double Categories

3 Products

Motivation – domains of definition via arrows

Set_p – category of sets and partial functions

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$\mathbf{Set}_p$ – category of sets and partial functions

Domain of definition

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Composition corresponds to intersection of the domains of definition.

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Composition corresponds to intersection of the domains of definition.

Note that if we call this assignment $\overline{(-)}$, then $f \leq g \iff f = g\overline{f}$

Definition (Restriction category)

A restriction category consists of a category $\mathcal{A}$ together with for every $A, B \in \mathcal{A}$, a map $\overline{(-)}: \mathcal{A}(A, B) \rightarrow \mathcal{A}(A, A)$ such that

- ① $f = f\overline{f}$
- ② $\overline{f}\overline{g} = \overline{g}\overline{f}$
- ③ $\overline{g}\overline{f} = \overline{gf}$
- ④ $f\overline{g}\overline{f} = \overline{g}f$

(whenever these equations make sense).

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(whenever these equations make sense).

Many consequences, e.g. $\overline{\text{id}_A} = \text{id}_A$, $\overline{f} \cdot \overline{f} = \overline{f}$, $\overline{\overline{f}} = \overline{f}$, ...

Examples

Example (Sets and partial functions)

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Example (Trivial restrictions)

For every category $\mathcal{C}$, we can construct a restriction category $\text{Triv}(\mathcal{C})$: category remains the same and for $f: C \rightarrow D$, we set $\overline{f} := \text{id}_C$.

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Example (Partical recursive functions)

Single object $\mathbb{N}$ and partial recursive functions as morphisms

$$\overline{f} = \text{id}_{\mathbb{N}}|_{\text{dom}(f)}$$

Example (Kleisli category for $- + 1$)

Let $\mathcal{C}$ be a distributive category (i.e. finite products, finite coproducts and $A \times (B + C) \cong A \times B + A \times C$), then the Kleisli category for the $- + 1$ monad can be equipped with the following restriction structure: For $f : A \rightarrow B + 1$, we set

$$\overline{f} := A \xrightarrow{\Delta} A \times A \xrightarrow{A \times f} A \times (B + 1) \cong A \times B + A \xrightarrow{\pi_1 + !} A + 1.$$

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Example (Stable lattices)

Let StabLat be the category, whose objects are meet-semilattices (i.e. posets with binary infima and top element) and morphisms are functions preserving binary infima (not necessarily the top element though!). Then StabLat^{op} has the following restriction structure: For $f \in \text{StabLat}(L, K)$, we set

$$\overline{f}(k) := k \wedge f(\top_L) : K \rightarrow K.$$

Examples – categories of partial maps

Definition (Stable system of monics)

A stable system of monics on a category $\mathcal{C}$ is a collection $\mathcal{M}$ of monomorphisms that

- is closed under compositions;
- contains all isomorphisms;
- is pullback-stable (pullbacks along $m \in \mathcal{M}$ exist and are in $\mathcal{M}$ again).

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Example

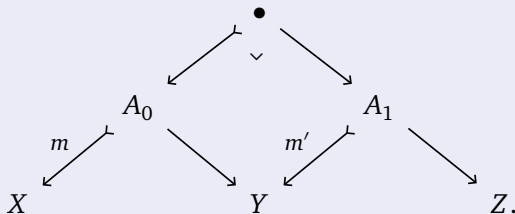
- All isomorphisms form a stable system of monics on every category.
- Monomorphisms in any category with pullbacks.
- Open subspaces of topological spaces.

Examples – Categories of partial maps

Definition (Category of partial maps)

Let $(\mathcal{C}, \mathcal{M})$ be a stable system of monics, we define a category $\text{Par}_{\mathcal{M}}(\mathcal{C})$, whose

- objects are those of $\mathcal{C}$;
- morphisms $\text{Par}_{\mathcal{M}}(\mathcal{C})(X, Y)$ are equivalence classes of spans $X \xleftarrow{m} A_0 \xrightarrow{f} Y$ with $m \in \mathcal{M}$;
- compositions:



Examples – Categories of partial maps

Example

The category of partial maps can be equipped with the structure of a restriction category with

$$\overline{X \overset{m}{\hookleftarrow} A_0 \xrightarrow{f} Y} := X \overset{m}{\hookleftarrow} A_0 \overset{m}{\twoheadrightarrow} X.$$

Definition (Total morphism)

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Total morphisms of a restriction category $\mathcal{C}$ form a wide subcategory of $\mathcal{C}$ denoted $\text{Total}(\mathcal{C})$.

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Several nice properties:

- Every restriction category is enriched in posets.
- $\bar{f} \leq \text{id}$
- If $f \leq g$ and f is total, then $f = g$.

Definition (Restriction functor)

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- For every restriction category $\mathcal{A}$, the inclusion $\text{Triv}(\text{Total}(\mathcal{A})) \rightarrow \mathcal{A}$ is a restriction functor.
- The inclusion of partial recursive functions into Set_p is a restriction functor.

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Proposition

Restriction functors preserve total morphisms and the order on hom sets.

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- objects $X, Y, Z, \dots$
- *tight* arrows between objects $f : X \rightarrow Z$
- *loose* arrows between objects $p : X \rightharpoonup Y$
- 2-cells of the form

$$\begin{array}{ccc} X & \xrightarrow{p} & Y \\ f \downarrow & \Downarrow \alpha & \downarrow g \\ Z & \xrightarrow{q} & W \end{array}$$

Definition of a double category

- we can compose such squares both in the loose and the tight direction

$$\begin{array}{ccccc}
 X & \xrightarrow{p} & Y & \xrightarrow{r} & V \\
 f \downarrow & \Downarrow \alpha & \downarrow g & \Downarrow \beta & \downarrow h \\
 Z & \xrightarrow{q} & W & \xrightarrow{s} & U
 \end{array}
 \mapsto
 \begin{array}{ccccc}
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 \mapsto
 \begin{array}{ccccc}
 X & \xrightarrow{p} & Y & & \\
 h \circ f \downarrow & \Downarrow \beta \circ \alpha & \downarrow k \circ g & & \\
 V & \xrightarrow{r} & U & &
 \end{array}$$

Both compositions are associative and compatible with each other:

$$(\delta \circ \gamma) \odot (\beta \circ \alpha) = (\delta \odot \beta) \circ (\gamma \odot \alpha).$$

Definition of a double category

- there are both tight $1_X : X \rightarrow X$ and loose $\text{id}_X : X \rightharpoonup X$ identities as well as 2-cell identities

$$\begin{array}{ccc}
 X & \xrightarrow{\text{id}_X} & X \\
 f \downarrow & \Downarrow 1_f & \downarrow f \\
 Y & \xrightarrow{\text{id}_Y} & Y
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with respect to the previous compositions.

Example (commutative squares)

From any category $\mathcal{C}$, we can form a double category in a trivial way by taking objects to be those of $\mathcal{C}$, loose and tight arrows to be the arrows of $\mathcal{C}$ and 2-cells to be commutative squares with the composition and identities inherited from $\mathcal{C}$.

Examples of double categories

Example (relations)

The double category $\mathbb{R}el(\mathbf{Set})$ of sets, functions and relations.

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$$\begin{array}{ccc} X & \xrightarrow{R} & Y \\ f \downarrow & \Downarrow & \downarrow g \\ Z & \xrightarrow{S} & W \end{array}$$

exactly when $x \sim_R y \implies f(x) \sim_S g(y)$.

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Remark. More generally, we can consider morphisms and relations in any *regular* category.

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$$\begin{array}{ccc} X & \begin{array}{c} \xrightarrow{L_1} \\ \perp \\ \xleftarrow{R_1} \end{array} & Y \\ f \downarrow & \Downarrow (\alpha_1, \alpha_2) & \downarrow g \\ Z & \begin{array}{c} \xrightarrow{L_2} \\ \perp \\ \xleftarrow{R_2} \end{array} & W \end{array}$$

is a pair of mate 2-cells in $\mathcal{C}$, i.e. $\alpha_1 : gL_1 \Rightarrow L_2f$ and $\alpha_2 : fR_1 \Rightarrow R_2g$.

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is a pair of mate 2-cells in $\mathcal{C}$, i.e. $\alpha_1 : gL_1 \Rightarrow L_2f$ and $\alpha_2 : fR_1 \Rightarrow R_2g$. Both compositions and identities are inherited from $\mathcal{C}$. Composition of adjunctions is an adjunction!

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$$\begin{array}{ccc} X & \xrightarrow{u} & Y \\ f \downarrow & \Downarrow & \downarrow g \\ Z & \xrightarrow{v} & W \end{array}$$

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- Both compositions and identities: those inherited from $\mathcal{A}$.

Examples of double restriction categories

Example ($\mathbb{R}(\text{Set}_p)$)

For the restriction category Set_p of sets and partial functions, the double category $\mathbb{R}(\text{Set}_p)$ has sets as objects, total functions as tight arrows and partial (not necessarily total) functions as loose arrows. A 2-cell exists if and only if the two composites of the square agree on where the top is defined.

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Example ($\mathbb{R}(\text{Par}_{\mathcal{M}}(\mathcal{C}))$)

Suppose $\mathcal{C}$ is a category with a stable system of monics $\mathcal{M}$ and form the category of partial arrows $\text{Par}_{\mathcal{M}}(\mathcal{C})$ with respect to $\mathcal{M}$. The induced double category has objects those of $\mathcal{C}$, loose arrows are spans of the form

$$X \overset{m}{\hookleftarrow} A_0 \overset{f}{\rightarrow} Y,$$

and tight arrows spans for which $m = 1_X$, i.e. plain arrows in $\mathcal{C}$.

Examples of double restriction categories

Example $(\mathbb{R}(\text{StabLat}^{\text{op}}))$

The double category $\mathbb{R}(\text{StabLat}^{\text{op}})$ has meet-semilattices as objects, stable maps as loose arrows and stable maps that preserve $\top$ as tight arrows.

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Example ($\mathbb{R}(\text{StabLat}^{\text{op}})$)

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exists if and only if

$$gu = vf\bar{u} \iff gu(x) = vf(x \wedge u(\top_x)) \quad \forall x \in X$$

Double functors

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to 2-cells

$$\begin{array}{ccc} FX & \xrightarrow{F\mu} & FY \\ Ff \downarrow & \Downarrow F\alpha & \downarrow Fg \\ FZ & \xrightarrow{Fv} & FW \end{array}$$

respecting compositions and identities.

Double restriction functors

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- on 2-cells:

$$\begin{array}{ccccc} X & \xrightarrow{u} & Y & & FX & \xrightarrow{F\mu} & FY \\ f \downarrow & \Downarrow \alpha & \downarrow g & \mapsto & Ff \downarrow & \Downarrow F\alpha & \downarrow Fg \\ Z & \xrightarrow{v} & W & & FZ & \xrightarrow{Fv} & FW \end{array}$$

($\mathbb{R}F$ sends the unique 2-cell to the unique 2-cell since F preserves ordering)

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Task: Find necessary and sufficient conditions.

1 Restriction Categories

2 Double Categories

3 Products

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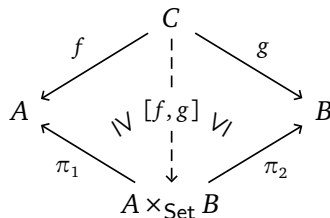
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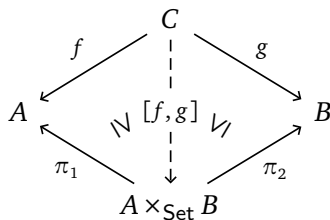
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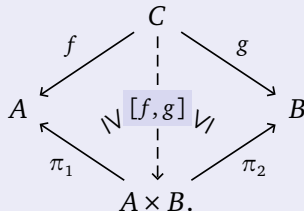


$[f, g]$ is the unique arrow satisfying the above inequalities and $\text{dom}([f, g]) = \text{dom}(f) \cap \text{dom}(g)$

Restriction products

Definition (Restriction products)

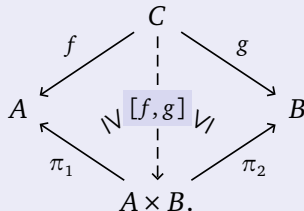
Let $\mathcal{A}$ be a restriction category and $A, B \in \mathcal{A}$. Then their restriction product consists of an object $A \times B$ together with two total projections $\pi_1: A \times B \rightarrow A$ and $\pi_2: A \times B \rightarrow B$ such that for every $f: C \rightarrow A, g: C \rightarrow B$ there exists a unique arrow $[f, g]: C \rightarrow A \times B$ such that $\overline{[f, g]} = \overline{f} \overline{g}$ and



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Restriction products give actual products in $Total(\mathcal{A})$.

Restriction products – examples

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More generally:

Example (Kleisli Category for $- + 1$)

Let $\mathcal{C}$ be a distributive category. Then the Kleisli (restriction) category Kl_{+1} has restriction products and these are precisely products in $\mathcal{C}$.

Double products

Definition (Double products)

A double category $\mathbb{D}$ has binary double products if for every $A, B \in \mathbb{D}$, there is an object $A \times B$ together with tight projections $\pi_1: A \times B \rightarrow A$, $\pi_2: A \times B \rightarrow B$ such that

- $A \times B, \pi_1, \pi_2$ form a product cone in the tight category of $\mathbb{D}$;
- for every loose arrows $h: A \rightharpoonup C, h': B \rightharpoonup D$, we have a loose arrow $h \times h': A \times B \rightharpoonup C \times D$ together with two squares

$$\begin{array}{ccc} A \times B & \xrightarrow{h \times h'} & C \times D \\ \pi_1 \downarrow & \Downarrow \Pi_1 & \downarrow \pi_1 \\ A & \xrightarrow{h} & C \end{array} \quad \begin{array}{ccc} A \times B & \xrightarrow{h \times h'} & C \times D \\ \pi_2 \downarrow & \Downarrow \Pi_2 & \downarrow \pi_2 \\ B & \xrightarrow{h'} & D \end{array}$$

such that $h \times h'$ with Π_1, Π_2 is a product cone in the category of loose arrows and squares.

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Example ($\mathbf{Rel}(\mathbf{Set})$)

The product of objects A, B is $A \times B$.

The product of relations $R \subseteq A \times C$ and $S \subseteq B \times D$ is the relation $R \times S \subseteq (A \times B) \times (C \times D)$ defined by

$$(a, b) \sim_{R \times S} (c, d) \iff a \sim_R c \text{ and } b \sim_S d.$$

The first horizontal projection Π_1 is the implication

$$(a, b) \sim_{R \times S} (c, d) \implies a \sim_R c.$$

Conjecture

A restriction category $\mathcal{A}$ has binary restriction products iff its associated double category $\mathbb{R}(\mathcal{A})$ has binary double products.

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We can similarly define restriction terminal objects and double terminal objects and conjecture this correspondence also for terminal objects.

- [1.] R. Cockett, S. Lack, *Restriction Categories I: categories of partial maps*, Theoretical Computer Science 270 (2002) 223-259.
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